Quantification: 682_z08
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PRE-VALIDATION: no number here has been checked against manual (ImageJ) counts.
PIXEL UNITS: lengths px, areas px^2, densities px^-1. The images carry no micron calibration.
Statistics are over interior lacunae only (those not touching the frame edge).

Per image
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measure	value
lacuna count	9
interior lacuna count	8
frame-edge lacuna count	1

Interior lacunae only: mean, median and sample SD (n = 8)
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measure	kind	mean	median	sd	units
area_px2	lacuna shape	2599.88	2671.5	1127.64	px^2
major_axis_length_px	lacuna shape	103.523	92.8874	29.9285	px
minor_axis_length_px	lacuna shape	33.6052	36.6597	6.7292	px
aspect_ratio	lacuna shape	3.087	2.9214	0.6438	unitless
eccentricity	lacuna shape	0.9401	0.9395	0.0221	unitless
solidity	lacuna shape	0.8296	0.873	0.1073	unitless
roots_count	headline	14.375	14	5.0973	count
ring_length_r30_px	headline	534.875	498	168.233	px
ring_length_r60_px	reported	1174.75	1086.5	387.866	px
owned_length_px	ownership-dependent	3561.83	2574.3	2128.04	px
edge_count	ownership-dependent	175.5	128.5	107.113	count
mean_edge_length_px	ownership-dependent	20.7739	20.4805	2.0162	px
perimeter_px	normalised	275.426	240.43	89.7432	px
ring_area_r30_px2	normalised	10379.6	9533.5	2334.04	px^2
ring_area_r60_px2	normalised	24427.5	24429.5	5136.87	px^2
in_frame_fraction_r30	normalised	1	1	0	unitless
in_frame_fraction_r60	normalised	0.9949	1	0.0144	unitless
ring_density_r30	normalised	0.0515015	0.0524088	0.0101109	px^-1
ring_density_r60	normalised	0.0481002	0.0476941	0.0106926	px^-1
roots_per_100px_perimeter	normalised	5.2326	5.4401	0.9413	per 100 px
ring_attached_length_r30_px	network v2	495.625	474.5	175.394	px
ring_attached_length_r60_px	network v2	1039.62	930	429.976	px
ring_length_w_r30_px	network v2	606.525	563.239	185.026	px
ring_length_w_r60_px	network v2	1330.56	1235.4	426.284	px
sholl_crossings_r10	network v2	13.125	11.5	5.0551	count
sholl_crossings_r20	network v2	15.5	14	3.2071	count
sholl_crossings_r30	network v2	16.875	14.5	5.0551	count
loop_count	loops, report only	17.875	12.5	12.9	count
loop_edge_share	loops, report only	0.3641	0.4122	0.1046	share
ring_width_mean_r30_px	width	7.1969	7.2603	0.3848	px

Per field
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measure	value	units
analysed_area_px2	1.02723e+06	px^2
total_skeleton_length_px	36075	px
canalicular_length_density_per_px	0.0351187	px^-1
junction_count	851	count
junction_density_per_px2	0.00082844	px^-2
lacuna_count	9	count

lacunae_per_px2	8.76e-06	px^-2
skeleton_component_count	266	count
mean_component_length_px	135.62	px
median_component_length_px	17.5	px
field_length_density_w_per_px	0.0395065	px^-1
field_density_without_flagged_per_px	0.0368076	px^-1
field_density_in_roi_per_px		px^-1
unattached_length_px	985	px
unattached_component_count	55	count
canalicular_length_density_attached_per_px	0.0351187	px^-1
field_length_density_with_unattached_per_px	0.0360776	px^-1
excluded_area_px2	0	px^2
analysed_field_area_px2	1.04858e+06	px^2
loop_count	179	count
loop_edge_share	0.3718	share
edge_count	1907	count

Canalicular width, the new measure

measure	value	units
width_mean_px	7.5791	px
width_median_px	8	px
width_p10_px	4	px
width_p90_px	10.198	px
width_pixels_used	34707	count

Canalicular width: 2 x the Euclidean distance transform of the canalicular mask at each skeleton pixel.
Measured only on skeleton pixels on the canalicular mask: the lacuna buffer and the
dilated vascular mask are excluded. The threads are about 2 to 6 px wide, so the width is close to the
pixel grid and quantised in steps of about 1 px, it moves with the threshold (a lower cut widens every
thread), and it has not been validated against any manual measurement.

Lacuna shape

lacuna_id	area_px2	major_axis_length_px	minor_axis_length_px	aspect_ratio	eccentricity	solidity	orientation_rad	centroid_row_px	centroid_col_px	on_border
1	2563	90.1026	36.7546	2.4515	0.913	0.9144	1.5381	342.687	87.595	False
2	3496	113.895	40.712	2.7976	0.9339	0.8552	-1.2503	434.254	701.917	False
3	4123	159.751	36.5648	4.369	0.9735	0.7865	1.4743	433.966	210.1	False
4	1521	89.609	30.1271	2.9744	0.9418	0.641	-1.2077	520.501	556.951	False
5	862	61.0771	21.2942	2.8683	0.9373	0.7077	-1.4443	545.484	438.164	False
6	547	50.4784	17.0874	2.9541	0.941	0.8116	-0.4973	672.987	1013.45	True
7	3601	127.976	37.0473	3.4544	0.9572	0.8909	1.4857	713.926	152.006	False
8	1853	92.0913	27.1748	3.3889	0.9555	0.9155	-1.2655	723.731	828.852	False
9	2780	93.6835	39.1668	2.3919	0.9084	0.926	-1.1692	908.613	727.092	False

Network per lacuna

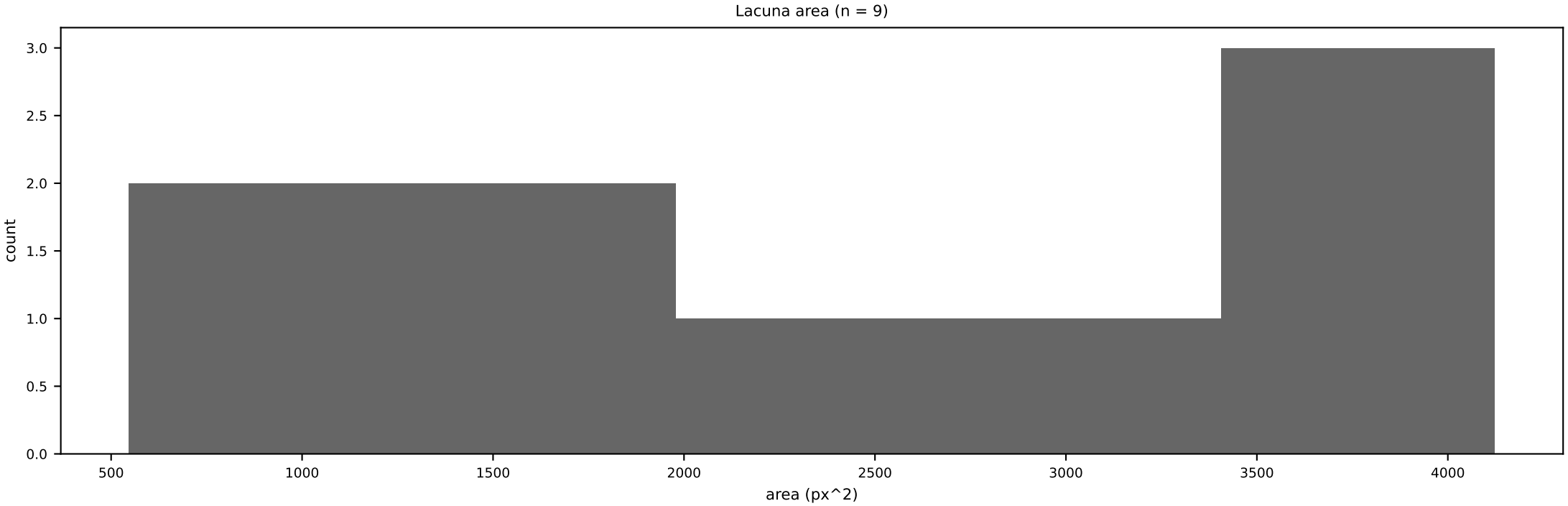
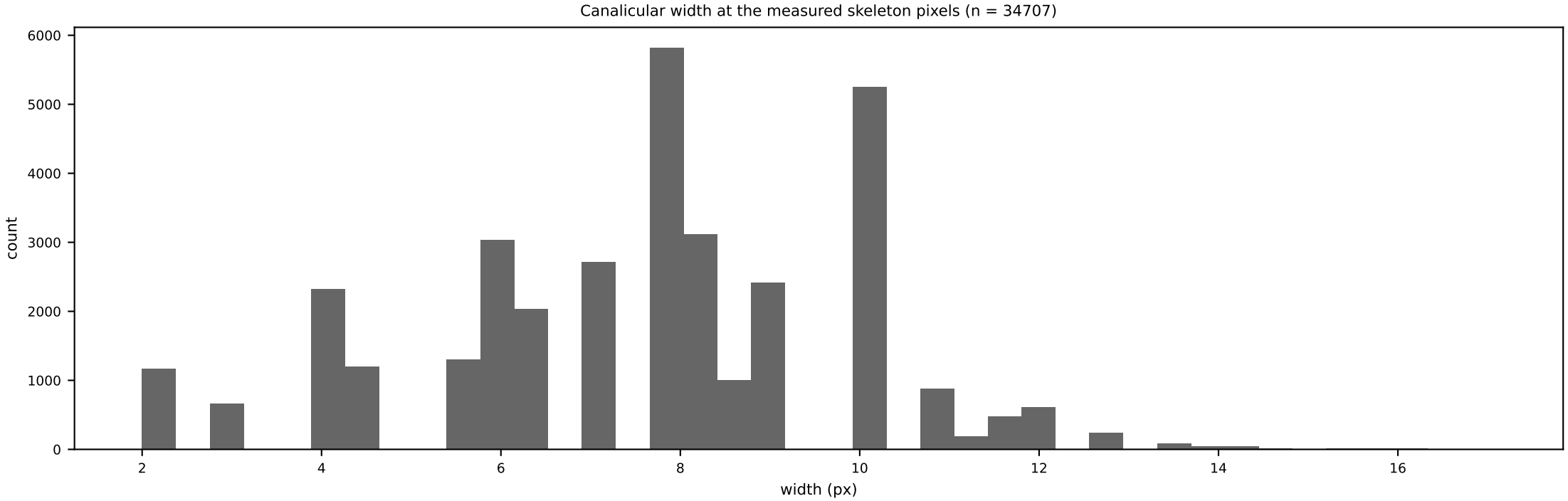
lacuna_id	roots_count	ring_length_r30_px	ring_length_r60_px	owned_length_px	edge_count	mean_edge_length_px	ring_width_mean_r30_px
1	14	516	1208	3986.83	215	18.5434	7.6974
2	18	496	1134	2967.73	151	19.6538	7.2847
3	23	872	1948	5935.65	324	18.3199	7.3441
4	11	458	869	2180.88	105	20.7703	7.1913
5	9	500	1039	2140.22	106	20.1907	6.3797
6	4	202	471	643.767	23	27.9899	6.1713
7	18	685	1516	7544.54	343	21.9957	7.4282
8	14	411	878	1789.29	79	22.6492	7.0137

9 8 341 806 1949.54 81 24.0684 7.236

Normalised per lacuna

lacuna_id	perimeter_px	ring_area_r30_px2	ring_area_r60_px2	in_frame_fraction_r30	in_frame_fraction_r60	ring_density_r30	ring_density_r60	roots_per_100px_perimeter
1	234.894	9374	20769	1	0.9593	0.0550459	0.0581636	5.9601
2	313.948	11274	26474	1	1	0.043995	0.0428345	5.7334
3	456.961	14983	32605	1	1	0.0581993	0.0597454	5.0333
4	233.238	9202	19621	1	1	0.0497718	0.0442893	4.7162
5	166.125	7387	17424	1	1	0.0676865	0.0596304	5.4176
6	122.604	3917	10590	0.6212	0.5802	0.0515701	0.0444759	3.2625
7	329.522	12020	29668	1	1	0.0569884	0.0510988	5.4625
8	222.752	9104	23858	1	1	0.045145	0.0368011	6.285
9	245.965	9693	25001	1	1	0.03518	0.0322387	3.2525

Network v2 per lacuna

lacuna_id	ring_attached_length_r30_px	ring_attached_length_r60_px	ring_length_w_r30_px	ring_length_w_r60_px	sholl_crossings_r10	sholl_crossings_r20	sholl_crossings_r30
1	501	1174	582.617	1354.59	13	14	17
2	477	978	568.73	1301.86	15	14	14
3	856	1897	976.968	2195.3	23	22	27
4	420	803	523.375	1003.58	9	14	13
5	472	882	557.747	1168.95	10	14	13
6	164	405	236.037	554.47	5	7	8
7	605	1343	775.47	1681.96	17	19	22
8	356	698	471.103	1019.36	10	14	15
9	278	542	396.191	918.879	8	13	14

682_z08: distributions, pre-validation, px